Protecting the force: Cybersecurity strategies for police

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[SYBScIT]
B.Sc. (Information Technology)
2025-2026

Under the Guidance of
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Mansi Bharat Gada Degree College of Commerce
Anjurphata, Bhiwandi.
Academic Year: 2025-2026

Oswal Shikshan & Rahat Sangh Sanchalit

MANSI BHARAT GADA DEGREE COLLEGE OFCOMMERCE

(Affiliated to University of Mumbai)
(NAAC Accredited 'B' Grade& ISO 9001:2015 certified)

CERTIFICATE

I hereby certify that Mr. Afaque Ansari, Student of **Mansi Bharat Gada Degree College of Commerce, with Roll Number:02**, has successfully completed a project titled **Protecting the Force :** in the subject of **Field Project of SYBScIT Semester III**, under the specialization of **Information Technology,** for the academic year 2025-2026.

To the best of my knowledge, the work submitted by the student is original, and the information included in the project is accurate and authentic.

Internal Guide Head of the Department Principal

DECLARATION

I, Mr. Ansari Afaque Md Shahnawaz Student of **Mansi Bharat Gada Degree College of Commerce** studying in **SYBScIT (Semester-III)** with **Roll Number**:02, hereby declare that I have completed the field project titled **Cybersecurity strategies for police** during the academic year 2025-2026.

I confirm that this project report is my own work. The information and data in this report are correct and have been collected and studied using **primary sources** as part of this project.

I have given proper credit to all sources used in this report by listing them in the **Appendix,** as per the given format.

Signature of the Student with Date
Afaque Ansari

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to everyone who supported and guided me in completing this project successfully.

First and foremost, I sincerely thank our **Principal, Mr. Mitesh Gosrani** of *Mansi Bharat Gada Degree College of Commerce*, for providing me with the opportunity and resources to carry out this project.

I extend my deep gratitude to our **Head of Department, Ms. Ansari Parveen from Information Technology** for her constant encouragement and valuable guidance throughout the academic year.

I am especially thankful to my **Internal Guide, Ms. Ansari Nusrat, Asst. Professor, Department of Information Technology**, for her continuous support, helpful suggestions, and encouragement at every stage of this project.

I also express my special thanks to my **parents and family members** for their love, motivation, and constant support, which helped me stay focused on my work.

My sincere thanks to my **classmates and friends** for their cooperation throughout this project journey.

Lastly, I would like to thank **all the individuals and organizations** who shared their time and valuable information, through which the data for this project was collected. Without their help, this project would not have been possible.

ABSTRACT

In today's digital world, the role of cybersecurity in policing has become increasingly critical. Police forces depend on advanced technologies to conduct investigations, store sensitive records, manage communication systems, and maintain law and order. However, this reliance on digital platforms exposes law enforcement agencies to serious cyber risks such as data breaches, ransomware attacks, identity theft, and unauthorized access to confidential information. A single cyber incident can compromise investigations, put officers and citizens at risk, and damage the credibility of the police.

This project focuses on strategies that police organizations can adopt to strengthen their cybersecurity frameworks. The study identifies key practices such as risk assessment, digital evidence protection, implementation of strong access controls, use of encryption tools, and development of incident response plans. In addition, it highlights the importance of regular training, cyber awareness programs, and collaboration with cyber cells and national security agencies to combat evolving threats. The project also emphasizes the use of advanced technologies like artificial intelligence, threat detection systems, and forensic tools to enhance the security of law enforcement operations.

The findings suggest that proactive and layered security measures are essential for safeguarding digital infrastructure in policing. Effective cybersecurity not only ensures data integrity and operational continuity but also helps maintain public trust in law enforcement. By adopting comprehensive cybersecurity strategies, police organizations can be better prepared to counter cyberattacks and uphold their responsibility of protecting society in both the physical and digital domains.

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CHAPTER 1: INTRODUCTION

In the digital age, police forces around the world are increasingly relying on technology to enhance public safety, streamline operations, and investigate crimes. From body-worn cameras and digital evidence management systems to crime databases and communication networks, law enforcement agencies handle vast amounts of sensitive data daily. This growing dependence on digital infrastructure exposes police departments to a wide range of cyber threats, including data breaches, ransomware attacks, and unauthorized access to critical systems.

Cybersecurity has therefore become a critical concern for modern policing. Ensuring the integrity, confidentiality, and availability of digital assets is essential not only for effective law enforcement but also for maintaining public trust. A single cybersecurity failure can compromise investigations, endanger officers, or expose the personal information of victims and suspects alike.

This paper explores key cybersecurity strategies tailored specifically for police organizations. It highlights the unique challenges they face in the cyber domain and outlines best practices for protecting law enforcement IT systems from evolving threats. By implementing robust cybersecurity frameworks, police departments can better safeguard their operations, protect sensitive data, and enhance their ability to serve and protect the public in an increasingly connected world.

CHAPTER 2: LITERATURE REVIEW

Since 2025

- ✓ Cybersecurity **and cybercrime combatting culture for african** police **services**
- ✓ L Loenen, J Jansen van Vuuren… - … Conference on Human …, 2025 - Springer

Since 2024

- ✓ **Partners in national** cyber security strategy**?: An analysis of** cyber security strategies **of Ministry of Defence and** police **in UK**
- ✓ CV Onuoha - 2024 - repository.londonmet.ac.uk

Since 2021

- ✓ Police **cybercrime training: Perceptions, pedagogy, and policy**
- ✓ T Cockcroft, M Shan-A-Kheda… - Policing: A Journal of …, 2021 - academic.oup.com

- ✓ **Facing cyberthreats in a crisis and post-crisis era: Rethinking security services response** strategy
- ✓ MJ Guitton, J Fréchette - Computers in human behaviour …, 2021 - pmc.ncbi.nlm.nih.gov

CHAPTER 3:METHODOLOGY

3.1 Research Method:

This study uses a Mixed Method Research approach, combining surveys for quantitative data (numbers) and open-ended questions for qualitative insights (opinions).

3.2 Tool Used:

1. Questionnaire – A structured set of questions prepared in Google Forms to collect both numerical data and opinions from students.
2. Interview – Online interview questions asked to students to gain deeper insights into their preferences.
3. Software/Apps – Tools like Google Forms for data collection and Excel for organizing and analyzing responses.

3.3 Sample Size:

The survey was completed by 35 college students, who participated in the study.

3.4 Sampling Method:

This study uses Non-Probability Sampling.

a. Convenience Sampling: We selected participants who were easy to reach, such as friends and classmates.

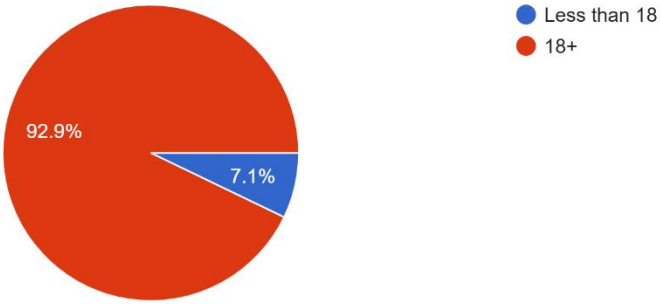
b. Purposive Sampling: We chose specific individuals who had knowledge about the topic.

c. Snowball Sampling: We asked participants to share the survey with others to increase responses

3.5 Data Analysis

Age

14 responses

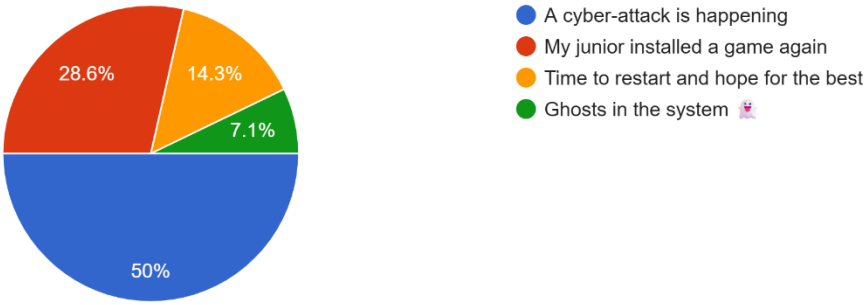


What best describes your occupation?

14 responses

If your computer starts acting strange, what's your first thought?

14 responses

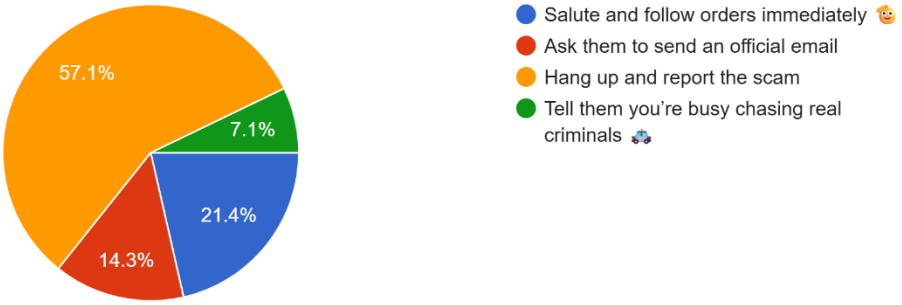


What's the best way to avoid cyber threats at work?

14 responses

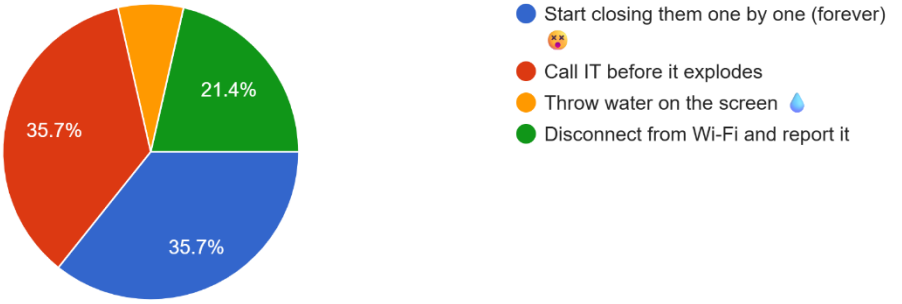
If your phone rings and the caller claims to be "Cyber Police Chief of the Internet," what should you do?

14 responses



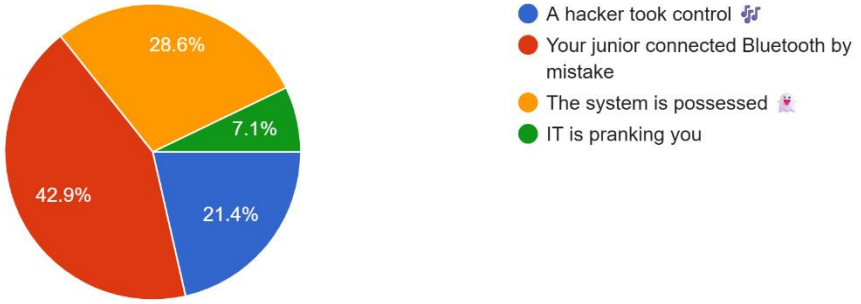
What's the safest reaction if your laptop suddenly shows 99 pop-ups?

14 responses



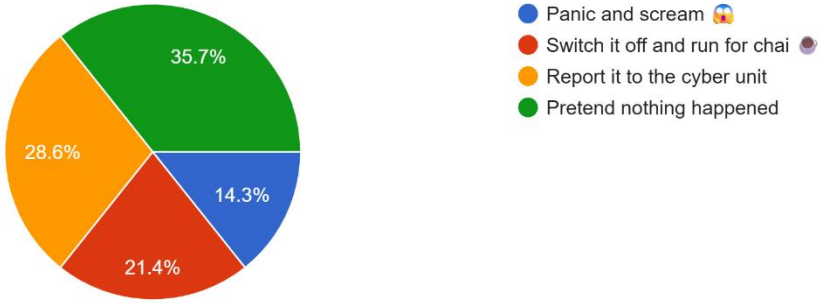
If your computer starts playing music by itself, what's most likely the reason?

14 responses



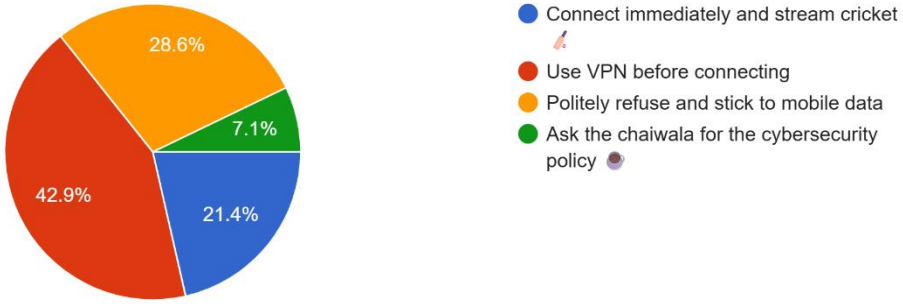
If your system suddenly shows "You've been hacked!" what's the first thing you do?

14 responses



If someone offers you free Wi-Fi at a tea stall near the station, what's the smart move?

14 responses



CHAPTER 4: FIELD WORK DESCRIPTION, OBSERVATION AND ANALYSIS

4.1 Field Work Description

During the field work on cybersecurity strategies for police, we visited the [insert police department/cybercrime unit] to observe and understand how law enforcement agencies protect sensitive information and prevent cybercrime. The visit began with a briefing by the cybercrime and IT team, where we learned about the types of cyber threats faced by police and the importance of maintaining robust cybersecurity measures.

4.1 Observations

During the field work, several key observations were made regarding the cybersecurity strategies employed by the police department. It was observed that the department uses a multi-layered security system, including firewalls, intrusion detection systems, and encryption, to protect sensitive data and communication channels. The personnel handling cybersecurity were highly trained and followed strict protocols for monitoring, detecting, and responding to cyber threats.

4.3 Analysis

The field work revealed that cybersecurity is a critical component of modern policing, as law enforcement agencies increasingly rely on digital systems to store sensitive data, communicate, and investigate crimes. The multi-layered security approach observed, including firewalls, intrusion detection systems, and encryption protocols, demonstrates a proactive strategy to prevent unauthorized access and cyberattacks.

CHAPTER 5: CONCLUSIONS AND RECOMMENDATIONS

5.1 Conclusion:

The field work on cybersecurity strategies for police provided valuable insights into how law enforcement agencies protect sensitive information and combat cybercrime. It was observed that an effective cybersecurity framework relies on a combination of advanced technology, trained personnel, strict protocols, and continuous monitoring. The use of firewalls, encryption, intrusion detection systems, and digital forensics tools ensures secure communication, data protection, and efficient investigation of cyber offenses. Collaboration between IT experts and police officers, adherence to legal and ethical standards, and regular updates to security systems are essential for addressing evolving cyber threats.

5.2 Recommendations:

1. **Regular Training for Personnel:** Conduct frequent cybersecurity awareness and skill-development programs for all police staff to handle evolving cyber threats effectively.
2. **Upgrade Security Infrastructure:** Continuously update firewalls, antivirus software, intrusion detection systems, and encryption protocols to address emerging vulnerabilities.
3. **Strengthen Digital Forensics Capabilities:** Invest in advanced tools and software for efficient collection, preservation, and analysis of digital evidence.
4. **Implement Strict Access Controls:** Ensure role-based access to sensitive data and maintain detailed logs of all digital transactions to prevent unauthorized access.
5. **Establish Incident Response Teams:** Form specialized teams for rapid detection, investigation, and mitigation of cyber incidents.
6. **Public Awareness Programs:** Educate citizens on cyber threats and safe online practices to reduce cases of cybercrime

REFERENCES

Journal Articles

1. **Thane Police Cyber Crime Investigation Cell**

- **Location:** 4th Floor, Office of Commissioner of Police, Near Kalwa Bridge, Thane (W)-400601
- **Phone:** 022-25429804
- **Email:** cp.thane.dcpeow@mahapolice.gov.in
- **Website:** Thane Police Cyber Cell

This unit investigates various cybercrimes, including online fraud, hacking, and digital harassment. They have been instrumental in addressing the growing concerns of cyber threats in the region.

2. **Thane Police Cyber Crime Investigation Cell**

- **Location:** 4th Floor, Office of Commissioner of Police, Near Kalwa Bridge, Thane (W)-400601
- **Phone:** 022-25429804
- **Email:** cp.thane.dcpeow@mahapolice.gov.in
- **Website:** Thane Police Cyber Cell

Established in 2005, the Cyber Crime Investigation Cell operates under the guidance of the Additional Commissioner of Police (Crime) and the Deputy Commissioner of Police (EOW). The cell investigates various types of cybercrimes, including email ID hacking, obscene emails, fake profiles, credit/debit card frauds, online lottery frauds, and SIM card frauds.

AI Tools Used

OpenAI:- https://chatgpt.com/

Google Scholar: https://scholar.google.com/

Google: https://google.com/

Google Scholar: https://gemini.ai/

APPENDICES

Appendix A: Guide Interaction Diary Form

I, the undersigned Mr. Ansari Afaque Md Shahnawaz Roll No. 02 studying in the Second Year of Bsc IT Fulltime Course is doing my project work under the guidance of Ms. Ansari Nusrat wish to state that I have met my Internal guide on the following dates mentioned below for Project Guidance:-

Sr.No.	Date	Signature of the Internal Guide
1	4th September 2025	
2	12th September 2025	
3	17th September 2025	
4	19th September 2025	
5	20th September 2025	
6	24th September 2025	
7	25th September 2025	

_____ _____

Signature of the Candidate Signature of Internal Guide

Appendix B: Google form Question

Protecting the force: Cybersecurity strategies for police

This form is part of a project on "Protecting the Force: Cybersecurity Strategies for Police." It aims to understand awareness, practices, and challenges in police cybersecurity. Your responses are confidential and will be used only for research purposes.

Sign in to Google to save your progress. Learn more

Full Name

Your answer

Your Email

Your answer

Age

- ◯ Less than 18
- ◯ 18+

What best describes your occupation?

- ◯ 👮‍♂️ Police officer (the real cyber warrior)
- ◯ 🎓 Student (future cyber warrior in training)
- ◯ 💼 Employee (fighting office emails & hackers together)
- ◯ 🏠 Homemaker (managing passwords and groceries alike)
- ◯ 📱 Tech enthusiast (just here for the fun)

Next

Clear form

If your computer starts acting strange, what's your first thought?

- ◯ A cyber-attack is happening
- ◯ My junior installed a game again
- ◯ Time to restart and hope for the best
- ◯ Ghosts in the system 👻

Which of these is the *worst* password for a police officer?

- ☐ password123
- ☐ donutlover 🍩
- ☐ IAmBatman 🦇
- ☐ MyNameAndBadgeNumber

If hackers stole the police Wi-Fi password, what's the most likely outcome?

- ◯ Netflix in the control room 📺
- ◯ Leaked case files 😱
- ◯ Slower internet (again!)
- ◯ Extra "guests" on CCTV cameras 👀

What do you usually do when asked to update your password?

- ◯ Make a brand-new strong password 💪
- ◯ Add “123” at the end of the old one 😅
- ◯ Forget about it until the system forces me
- ◯ Ask IT to reset it (again)

If a hacker pretends to be your boss in an email, how can you tell it’s fake?

- ◯ The email is full of spelling mistakes
- ◯ They ask for gift cards 🎁
- ◯ They suddenly start being polite
- ◯ All of the above

What’s the best way to avoid cyber threats at work?

- ◯ Stay alert and follow protocols
- ◯ Have a strong password policy
- ◯ Never click suspicious links
- ◯ Pull the plug on the computer and go for tea ☕

What's the safest reaction if your laptop suddenly shows 99 pop-ups?

- ◯ Start closing them one by one (forever) 😵
- ◯ Call IT before it explodes
- ◯ Throw water on the screen 💧
- ◯ Disconnect from Wi-Fi and report it

If someone offers you free Wi-Fi at a tea stall near the station, what's the smart move?

- ◯ Connect immediately and stream cricket 🏏
- ◯ Use VPN before connecting
- ◯ Politely refuse and stick to mobile data
- ◯ Ask the chaiwala for the cybersecurity policy ☕

If your computer starts playing music by itself, what's most likely the reason?

- ◯ A hacker took control 🎶
- ◯ Your junior connected Bluetooth by mistake
- ◯ The system is possessed 👻
- ◯ IT is pranking you

If your computer starts playing music by itself, what's most likely the reason?

- ◯ A hacker took control 🎶
- ◯ Your junior connected Bluetooth by mistake
- ◯ The system is possessed 👻
- ◯ IT is pranking you

If a hacker offered you free samosas in exchange for your password, what would you do?

- ◯ Laugh and walk away 😄
- ◯ Say "Nice try, buddy"
- ◯ Actually consider it (because samosas 🥟)
- ◯ Report the attempt immediately

If your system suddenly shows "You've been hacked!" what's the first thing you do?

- ◯ Panic and scream 😱
- ◯ Switch it off and run for chai ☕
- ◯ Report it to the cyber unit
- ◯ Pretend nothing happened

If a hacker offered you free samosas in exchange for your password, what would you do?

- ◯ Laugh and walk away 😄
- ◯ Say "Nice try, buddy"
- ◯ Actually consider it (because samosas 🥟)
- ◯ Report the attempt immediately

If your system suddenly shows "You've been hacked!" what's the first thing you do?

- ◯ Panic and scream 😱
- ◯ Switch it off and run for chai 🍵
- ◯ Report it to the cyber unit
- ◯ Pretend nothing happened

👮‍♀️ ⚡ "Protecting the Force" achieved — your answers are more secure than your Wi-Fi password.

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

Back Submit Clear form

Appendix C: Sample Data(Excel Table)

Sr.no	Timestamp	Full Name
1	16/09/2025 12:32:34	Waqas Khan
\2	16/09/2025 12:32:34	Farid Ansari
3	16/09/2025 12:33:47	Ridhu
4	16/09/2025 12:49:15	Rahul Patil
5	16/09/2025 13:04:51	Sonali Ashok Mali
6	16/09/2025 13:16:35	Manas Papua
7	16/09/2025 13:33:25	Ashok Mali
8	16/09/2025 14:02:41	HRISHIKESH JHA
9	16/09/2025 14:14:46	Urvi ankle
10	16/09/2025 15:14:31	Wasif Syed
11	16/09/2025 17:17:21	Nadeem Choudhary
12	16/09/2025 17:18:44	Roshni Gupta
13	16/09/2025 18:18:49	Avanthika Chemmani
14	16/09/2025 20:09:56	Ujwal Singh
15	16/09/2025 20:13:35	Prince Kumar
16	16/09/2025 20:31:52	Ali shaikh
17	16/09/2025 20:50:47	Rehan Gauri
18	16/09/2025 20:57:55	Krish Patil
19	16/09/2025 21:00:24	Kaveri Yamara
20	16/09/2025 21:06:30	Jaiswal Roshni

Student Feedback on FP

Student Name: Ansari Afaque Md Shahnawaz Ansari

Seat No. /Roll No.:02

Email: afaqueansari707@gmail.com

Department: B.Sc. (Information Technology)

Name of the Mentor: Ms.Ansari Nusrat

Title/Heading of Field Project:

Brief description of FP carried out: During the visit, we were given a detailed briefing by the officers of the Cyber Crime Investigation Cell. They explained the types of cyber threats commonly faced by the police, such as hacking, online frauds, phishing attacks, and digital harassment. The officers demonstrated the use of firewalls, intrusion detection systems, and encryption methods employed to secure communication and sensitive information.

We also observed the **digital forensics process**, including how evidence is collected, preserved, and analysed during cybercrime investigations. Officers highlighted the importance of adhering to legal and ethical guidelines while handling digital evidence. Additionally, the team explained the role of specialized initiatives, such as the **Cryptocurrency Investigation Cell** and the **WhatsApp chatbot for citizen services**, which help the police respond efficiently to cyber complaints and enhance public interaction.

Was your internship experience related to your major area of study?

- Yes, to a large degree
- Yes, to a slight degree
- No, not related at all

Indicate the degree to which you agree or disagree with the following statements.

This experience has:	Strongly Agree	Agree	No opinion	Disagree	Strongly Disagree
Given me the opportunity to explore a career field		✓			
Allowed me to apply classroom theory to practice	✓				
Helped me develop my decisionmaking and problem-		✓			
Expanded my knowledge about the work world before permanent employment	✓				
Helped me develop my written and oral communication skills	✓				
Provided a chance to use leadership skills (influence others, develop ideas with others, stimulate		✓			
Expanded my sensitivity to the ethical implications of the work involved		✓			
Made it possible for me to be more confident in new situations			✓		
Given me a chance to improve my interpersonal skills		✓			
Helped me learn to handle responsibility and use my time wisely					
Helped me discover new aspects of myself that I didn't know existed before			✓		
Helped me develop new interests and abilities	✓				
Helped me clarify my career goals			✓		
Allowed me to acquire information and/ or use equipment not available at my Institute	✓				
Allowed me to realize socio-economic issues in the society		✓			

❖ **In the Institute FP program, faculty members are expected to be mentors for students Do you feel that your faculty mentor served such a function? Why or why not?**

Ans. Yes, my faculty mentor was supportive and accessible. They offered guidance, regular feedback, and encouraged my growth throughout the FP.

❖ **How well were you able to accomplish the initial goals, tasks and new skills that were set down in your learning contract? In what ways were you able to take a new direction or expand beyond your contract? Why were some goals not accomplished adequately?**

Ans. Most goals were achieved, including gaining new skills and completing key tasks. I also explored new areas beyond the contract. Some goals weren't fully met due to time limits and resource constraints.

❖ **In what areas did you most develop and improve?**

Ans. I improved most in independent research, critical thinking, and communication. I also became more confident using relevant tools and methods.

❖ **What has been the most significant accomplishment or satisfying moment of your FP?**

Ans. Completing a major milestone and receiving positive feedback from my mentor was the most rewarding part of the experience.

❖ **What did you dislike about the FP?**

Ans. At times, the lack of structure and limited resources were challenging, but they also helped me grow in adaptability and problem-solving.

❖ **Considering your overall experience, how would you rate this FP? (Circle one). – Satisfactory/ Good/ Excellent**

❖ **Give suggestions as to how your FP experience could have been improved. (Could you have handled added responsibility? Would you have liked more discussions with your professor concerning your FP? Was closer supervision needed? Was more of an orientation required?)**

The FP could be improved by giving students more time for fieldwork, adding regular progress discussions with mentors, and providing sample projects as references for better clarity.

Signature of Student :

Name: Afaque Ansari

Date: